10. $Q = It$

$= 1A \times 60S$

$96500\ C\ \text{charge} \equiv 6.022 \times 10^{23}\ \text{electrons}$

$60\ C\ \text{charge} \equiv \frac{6.022 \times 10^{23}}{96500} \times 60$

$= 3.744 \times 10^{20}\ \text{electrons}$

(Option (C))

11. In general, specific conductance of an electrolyte decreases with dilution. So, 0.002N solution has least specific conductance.

(Option (b))

12. Charging : anode : $PbSO_4(s) + 2e^- \rightarrow Pb(s) + SO_4^{-2}(aq)$

Cathode : $PbSO_4(s) + 2H_2O(l) \rightarrow PbO_2(s) + SO_4^{-2}(aq) + 2e^-$

(Option (C))

13. Option (a) I and IV

14. $E^o_{Zn^{2+}|Zn} = -0.76V$ and $E^o_{Fe^{2+}|Fe} = -0.44V$ Zinc has higher negative electrode potential than iron, iron cannot be coated on zinc.

Option (d)

15. Both are false

i) Dry air has no reaction with iron

ii) Rust has the composition $Fe_2O_3 . x\ H_2O$

(Option (d))

16. (Option (a))

17. $\alpha = \frac{\Lambda}{\Lambda_o} = \frac{6}{400}$

$K_a = \alpha^2 C$

$= \frac{6}{400} \times \frac{6}{400} \times \frac{1}{36}$

$= 6.25 \times 10^{-6}$

Option (b)

18. $R = \rho . \frac{l}{A}$

$\text{cell constant} = \frac{R}{\rho}$

$= \kappa . R \left(\frac{1}{\rho} = \kappa \right)$

$= 1.25 \times 10^{-3}\ \Omega^{-1} cm^{-1} \times 800\ \Omega$

$= 1\ cm^{-1}$

Option (c)

19. Option (d)

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